

## ACKNOWLEDGMENTS

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## REFERENCES

- Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- AI@Meta. Llama 3 model card, 2024. URL [https://github.com/meta-llama/llama3/blob/main/MODEL\\_CARD.md](https://github.com/meta-llama/llama3/blob/main/MODEL_CARD.md).
- Noga Alon. Problems and results in extremal combinatorics—i. *Discrete Mathematics*, 273(1-3): 31–53, 2003.
- Anthropic. The claude 3 model family: Opus, sonnet, haiku. Technical report, Anthropic, 2024.
- Nora Belrose, Zach Furman, Logan Smith, Danny Halawi, Igor Ostrovsky, Lev McKinney, Stella Biderman, and Jacob Steinhardt. Eliciting latent predictions from transformers with the tuned lens. *arXiv preprint arXiv:2303.08112*, 2023.
- Sid Black, Lee Sharkey, Leo Grinsztajn, Eric Winsor, Dan Braun, Jacob Merizian, Kip Parker, Carlos Ramón Guevara, Beren Millidge, Gabriel Alfour, et al. Interpreting neural networks through the polytope lens. *arXiv preprint arXiv:2211.12312*, 2022.
- Joseph Bloom. Open source sparse autoencoders for all residual stream layers of gpt2 small. <https://www.alignmentforum.org/posts/f9EgflSurAiqRJySD/open-source-sparse-autoencoders-for-all-residual-stream>, 2024.
- Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023. <https://transformer-circuits.pub/2023/monosemantic-features/index.html>.
- Sébastien Bubeck, Varun Chandrasekaran, Ronen Eldan, Johannes Gehrke, Eric Horvitz, Ece Kamar, Peter Lee, Yin Tat Lee, Yuanzhi Li, Scott Lundberg, et al. Sparks of artificial general intelligence: Early experiments with gpt-4. *arXiv preprint arXiv:2303.12712*, 2023.
- Arthur Conmy, Augustine Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. Towards automated circuit discovery for mechanistic interpretability. *Advances in Neural Information Processing Systems*, 36:16318–16352, 2023.
- Róbert Csordás, Christopher Potts, Christopher D Manning, and Atticus Geiger. Recurrent neural networks learn to store and generate sequences using non-linear representations. *arXiv preprint arXiv:2408.10920*, 2024.
- Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. *arXiv preprint arXiv:2309.08600*, 2023.
- David Dalrymple, Joar Skalse, Yoshua Bengio, Stuart Russell, Max Tegmark, Sanjit Seshia, Steve Omohundro, Christian Szegedy, Ben Goldhaber, Nora Ammann, et al. Towards guaranteed safe ai: A framework for ensuring robust and reliable ai systems. *arXiv preprint arXiv:2405.06624*, 2024.

- Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, Roger Grosse, Sam McCandlish, Jared Kaplan, Dario Amodei, Martin Wattenberg, and Christopher Olah. Toy models of superposition. *Transformer Circuits Thread*, 2022. [https://transformer-circuits.pub/2022/toy\\_model/index.html](https://transformer-circuits.pub/2022/toy_model/index.html).
- Simon Foucart and Holger Rauhut. *A Mathematical Introduction to Compressive Sensing*. 2013.
- Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, Horace He, Anish Thite, Noa Nabeshima, et al. The pile: An 800gb dataset of diverse text for language modeling. *arXiv preprint arXiv:2101.00027*, 2020.
- Semyon Aranovich Gershgorin. über die abgrenzung der eigenwerte einer matrix. *Izvestiya Rossijskoi akademii nauk. Seriya matematicheskaya*, (6):749–754, 1931.
- Rhys Gould, Euan Ong, George Ogden, and Arthur Conmy. Successor heads: Recurring, interpretable attention heads in the wild. *arXiv preprint arXiv:2312.09230*, 2023.
- Wes Gurnee and Max Tegmark. Language models represent space and time. *arXiv preprint arXiv:2310.02207*, 2023.
- Michael Hanna, Ollie Liu, and Alexandre Variengien. How does gpt-2 compute greater-than?: Interpreting mathematical abilities in a pre-trained language model. *Advances in Neural Information Processing Systems*, 36, 2024.
- Benjamin Heinzerling and Kentaro Inui. Monotonic representation of numeric properties in language models. *arXiv preprint arXiv:2403.10381*, 2024.
- Nicholas J. Higham. Singular value inequalities. <https://nhigham.com/2021/05/04/singular-value-inequalities/>, May 2021.
- Bill Johnson ([https://mathoverflow.net/users/2554/bill\\_johnson](https://mathoverflow.net/users/2554/bill_johnson)). Almost orthogonal vectors. MathOverflow. URL <https://mathoverflow.net/q/24873>. URL:<https://mathoverflow.net/q/24873> (version: 2010-05-16).
- Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. Mistral 7b. *arXiv preprint arXiv:2310.06825*, 2023.
- Yibo Jiang, Goutham Rajendran, Pradeep Ravikumar, Bryon Aragam, and Victor Veitch. On the origins of linear representations in large language models. *arXiv preprint arXiv:2403.03867*, 2024.
- William B. Johnson and Joram Lindenstrauss. Extensions of lipschitz mappings into a hilbert space. In *Conference in modern analysis and probability (New Haven, Conn., 1982)*, volume 26 of *Contemporary Mathematics*, pp. 189–206. American Mathematical Society, Providence, RI, 1984. ISBN 0-8218-5030-X. doi: 10.1090/conm/026/737400.
- Subhash Kantamneni and Max Tegmark. Language models use trigonometry to do addition. *arXiv preprint arXiv:2502.00873*, 2025.
- Silvio Lattanzi, Thomas Lavastida, Kefu Lu, and Benjamin Moseley. A framework for parallelizing hierarchical clustering methods. In *Machine Learning and Knowledge Discovery in Databases: European Conference, ECML PKDD 2019, Würzburg, Germany, September 16–20, 2019, Proceedings, Part I*, pp. 73–89. Springer, 2020.
- Kenneth Li, Aspen K Hopkins, David Bau, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Emergent world representations: Exploring a sequence model trained on a synthetic task. *arXiv preprint arXiv:2210.13382*, 2022.
- Ziming Liu, Ouail Kitouni, Niklas S Nolte, Eric Michaud, Max Tegmark, and Mike Williams. Towards understanding grokking: An effective theory of representation learning. *Advances in Neural Information Processing Systems*, 35:34651–34663, 2022.

- Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. *arXiv preprint arXiv:2310.06824*, 2023.
- Samuel Marks, Can Rager, Eric J Michaud, Yonatan Belinkov, David Bau, and Aaron Mueller. Sparse feature circuits: Discovering and editing interpretable causal graphs in language models. *arXiv preprint arXiv:2403.19647*, 2024.
- Jake Mendel. Sae feature geometry is outside the superposition hypothesis. *AI Alignment Forum*, 2024. URL <https://www.alignmentforum.org/posts/MFBTjb2qf3ziWmzz6/sae-feature-geometry-is-outside-the-superposition-hypothesis>.
- Eric J Michaud, Isaac Liao, Vedang Lad, Ziming Liu, Anish Mudide, Chloe Loughridge, Zifan Carl Guo, Tara Rezaei Kheirkhah, Mateja Vukelić, and Max Tegmark. Opening the ai black box: program synthesis via mechanistic interpretability. *arXiv preprint arXiv:2402.05110*, 2024.
- Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg S Corrado, and Jeff Dean. Distributed representations of words and phrases and their compositionality. *Advances in neural information processing systems*, 26, 2013a.
- Tomáš Mikolov, Wen-tau Yih, and Geoffrey Zweig. Linguistic regularities in continuous space word representations. In *Proceedings of the 2013 conference of the north american chapter of the association for computational linguistics: Human language technologies*, pp. 746–751, 2013b.
- Depen Morwani, Benjamin L Edelman, Costin-Andrei Oncescu, Rosie Zhao, and Sham Kakade. Feature emergence via margin maximization: case studies in algebraic tasks. *arXiv preprint arXiv:2311.07568*, 2023.
- Aaron Mueller, Jannik Brinkmann, Millicent Li, Samuel Marks, Koyena Pal, Nikhil Prakash, Can Rager, Aruna Sankaranarayanan, Arnab Sen Sharma, Jiuding Sun, et al. The quest for the right mediator: A history, survey, and theoretical grounding of causal interpretability. *arXiv preprint arXiv:2408.01416*, 2024.
- Neel Nanda and Joseph Bloom. Transformerlens. <https://github.com/TransformerLensOrg/TransformerLens>, 2022.
- Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. *arXiv preprint arXiv:2301.05217*, 2023a.
- Neel Nanda, Andrew Lee, and Martin Wattenberg. Emergent linear representations in world models of self-supervised sequence models. *arXiv preprint arXiv:2309.00941*, 2023b.
- Chris Olah. What is a linear representation? what is a multidimensional feature? *Transformer Circuits Thread*, 2024. URL <https://transformer-circuits.pub/2024/july-update/index.html#linear-representations>.
- Chris Olah, Nick Cammarata, Ludwig Schubert, Gabriel Goh, Michael Petrov, and Shan Carter. Zoom in: An introduction to circuits. *Distill*, 2020. doi: 10.23915/distill.00024.001. <https://distill.pub/2020/circuits/zoom-in>.
- Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. *arXiv preprint arXiv:2311.03658*, 2023.
- Baolin Peng, Chunyuan Li, Pengcheng He, Michel Galley, and Jianfeng Gao. Instruction tuning with gpt-4. *arXiv preprint arXiv:2304.03277*, 2023.
- Jeffrey Pennington, Richard Socher, and Christopher D Manning. Glove: Global vectors for word representation. In *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)*, pp. 1532–1543, 2014.
- Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever, et al. Language models are unsupervised multitask learners. *OpenAI blog*, 2019.

- Adam Shai, Paul Riechers, Lucas Teixeira, Alexander Oldenziel, and Sarah Marzen. Transformers represent belief state geometry in their residual stream. <https://www.alignmentforum.org/posts/gTZ2SxesbHckJ3CkF/transformers-represent-belief-state-geometry-in-their>, 2024.
- Alessandro Stolfo, Yonatan Belinkov, and Mrinmaya Sachan. A mechanistic interpretation of arithmetic reasoning in language models using causal mediation analysis. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 7035–7052, 2023.
- Aaquib Syed, Can Rager, and Arthur Conmy. Attribution patching outperforms automated circuit discovery. *arXiv preprint arXiv:2310.10348*, 2023.
- Gemini Team, Rohan Anil, Sebastian Borgeaud, Yonghui Wu, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023.
- Max Tegmark and Steve Omohundro. Provably safe systems: the only path to controllable agi. *arXiv preprint arXiv:2309.01933*, 2023.
- Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: a circuit for indirect object identification in gpt-2 small. *arXiv preprint arXiv:2211.00593*, 2022.
- Adam Yedidia. Gpt-2’s positional embedding matrix is a helix, 2023a. URL <https://www.lesswrong.com/posts/qvWP3aBDBaqXvPNhS/gpt-2-s-positional-embedding-matrix-is-a-helix>. Accessed: 2024-09-16.
- Adam Yedidia. The positional embedding matrix and previous-token heads: how do they actually work?, 2023b. URL <https://www.lesswrong.com/posts/zRA8B2FJLtTYRgie6/the-positional-embedding-matrix-and-previous-token-heads-how>. Accessed: 2024-09-17.
- Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models: Metrics and methods. *arXiv preprint arXiv:2309.16042*, 2023.
- Ziqian Zhong, Ziming Liu, Max Tegmark, and Jacob Andreas. The clock and the pizza: Two stories in mechanistic explanation of neural networks. *Advances in Neural Information Processing Systems*, 36, 2024.

## A MULTI-DIMENSIONAL FEATURE CAPACITY

The Johnson-Lindenstrauss (JL) Lemma (Johnson & Lindenstrauss, 1984) implies that we can choose  $e^{Cd\delta^2}$  pairwise one-dimensional  $\delta$ -orthogonal vectors to satisfy Hypothesis 1 for some constant  $C$ , thus allowing us to build the model’s feature space with a number of one-dimensional  $\delta$ -orthogonal features exponential in  $d$ . We now prove a similar result for low-dimensional projections (the main idea of the proof is to combine  $\delta$ -orthogonal vectors as guaranteed from the JL lemma):

**Theorem 1.** *For any  $d'$  and  $\delta$ , it is possible to choose  $\frac{1}{d_{\max}}e^{C_1(d/d'^2)\delta^2}$  pairwise  $\delta$ -orthogonal matrices  $\mathbf{A}_i \in \mathbb{R}^{n_i \times d'}$  for some constant  $C_1$ . Furthermore, it is not possible to choose more than  $e^{C_2(d-d_{\max}\delta \log(\frac{1}{\delta}))}$  for some constant  $C_2$ .*

We will first prove a lemma that will help us prove Theorem 1.

**Lemma 1.** *Pick  $n$  pairwise  $\delta$ -orthogonal unit vectors in  $\mathbf{v}_1, \dots, \mathbf{v}_n \in \mathbb{R}^d$ . Let  $\mathbf{y} \in \mathbb{R}^d$  be a unit norm vector that is a linear combination of unit norm vectors  $\mathbf{v}_1, \dots, \mathbf{v}_n$  with coefficients  $z_1, \dots, z_n \in \mathbb{R}$ . We can write  $\mathbf{A} = [\mathbf{v}_1, \dots, \mathbf{v}_n]$  and  $\mathbf{z} = [z_1, \dots, z_n]^T$ , so that we have  $\mathbf{y} = \sum_{k=1}^n z_k \mathbf{v}_k = \mathbf{A}\mathbf{z}^T$  with  $\|\mathbf{y}\|_2 = 1$ . Then,*

$$\left| \sum_{k=1}^n \mathbf{z}_k \right| = \|\mathbf{z}\|_1 \leq \sqrt{\frac{n}{1-\delta n}}$$

*Proof.* We will first bound the  $L_2$  norm of  $\mathbf{z}$ . If  $\sigma_n$  is the minimum singular value of  $\mathbf{A}$ , then we have via standard singular value inequalities (Higham, 2021)

$$\sigma_n \leq \frac{\|\mathbf{y}\|_2}{\|\mathbf{z}\|_2} \implies \|\mathbf{z}\|_2 \leq \frac{\|\mathbf{y}\|_2}{\sigma_n} = \frac{1}{\sigma_n}$$

Thus we now lower bound  $\sigma_n$ . The singular values are the square roots of the eigenvalues of the matrix  $\mathbf{A}^T \mathbf{A}$ , so we now examine  $\mathbf{A}^T \mathbf{A}$ . Since all elements of  $\mathbf{A}$  are unit vectors, the diagonal of  $\mathbf{A}^T \mathbf{A}$  is all ones. The off diagonal elements are dot products of pairs of  $\delta$ -orthogonal vectors, and so are within the range  $[-\delta, \delta]$ . Then by the Gershgorin circle theorem (Gershgorin, 1931), all eigenvalues  $\lambda_i$  of  $\mathbf{A}^T \mathbf{A}$  are in the range

$$(1 - \delta(n-1), 1 + \delta(n-1))$$

In particular,  $\sigma_n^2 = \lambda_n \geq 1 - \delta(n-1)$ , and thus  $\sigma_n \geq \sqrt{1 - \delta(n-1)}$ . Plugging into our upper bound for  $\|\mathbf{z}\|_2$ , we have that  $\|\mathbf{z}\|_2 \leq 1/\sqrt{1 - \delta(n-1)}$ . Finally, the largest  $L_1$  for a point on an  $n$ -hypersphere of radius  $r$  is when all dimensions are equal and such a point has magnitude  $\sqrt{nr}$ , so

$$\|\mathbf{z}\|_1 \leq \sqrt{\frac{n}{1 - \delta(n-1)}} \leq \sqrt{\frac{n}{1 - \delta n}}$$

□

**Theorem 1.** *For any  $d'$  and  $\delta$ , it is possible to choose  $\frac{1}{d_{\max}}e^{C_1(d/d'^2)\delta^2}$  pairwise  $\delta$ -orthogonal matrices  $\mathbf{A}_i \in \mathbb{R}^{n_i \times d'}$  for some constant  $C_1$ . Furthermore, it is not possible to choose more than  $e^{C_2(d-d_{\max}\delta \log(\frac{1}{\delta}))}$  for some constant  $C_2$ .*

*Proof.* By the JL lemma (Johnson & Lindenstrauss, 1984; <https://mathoverflow.net/users/2554/bill-johnson>), for any  $d$  and  $\delta$ , we can choose  $e^{Cd\delta^2}$   $\delta$ -orthogonal unit vectors in  $\mathbb{R}^d$  indexed as  $\mathbf{v}_i$ , for some constant  $C$ . Let  $\mathbf{A}_i = [\mathbf{v}_{d_{\max}*i}, \dots, \mathbf{v}_{d_{\max}*i+n_i-1}]$  where each element in the brackets is a column. Then by construction all  $\mathbf{A}_i$  are matrices composed of unique  $\delta$ -orthogonal vectors and there are  $\frac{1}{d_{\max}}e^{Cd\delta^2}$  matrices  $\mathbf{A}_i$ .

Now, consider two of these matrices  $\mathbf{A}_i = [\mathbf{v}_1, \dots, \mathbf{v}_{n_i}]$  and  $\mathbf{A}_j = [\mathbf{u}_1, \dots, \mathbf{u}_{n_j}]$ ,  $i \neq j$ ; we will prove that they are  $f(\delta)$ -orthogonal for some function  $f$ . Let  $\mathbf{y}_i = \sum_{k=1}^{n_i} z_{i,k} \mathbf{v}_k$  be a vector in the colspace of  $\mathbf{A}_i$  and  $\mathbf{y}_j = \sum_{k=1}^{n_j} z_{j,k} \mathbf{u}_k$  be a vector in the colspace of  $\mathbf{A}_j$ , such that  $\mathbf{y}_i$  and  $\mathbf{y}_j$  are unit vectors. To prove  $f(\delta)$ -orthogonality, we must bound the absolute dot product between  $\mathbf{y}_i$  and  $\mathbf{y}_j$ :